

PROJECT DOCUMENTATION

MERN STACK BOOK STORE

System Design, Implementation & Verification Walkthrough

DEVELOPMENT TEAM & ACADEMIC ROLES:

1. Roopesh Araja - Technical Architecture, Entity Relation Diagram, Features, Roles
2. Palla Satya Ramanaidu - MVC Pattern, Creating Project Folder, Client Setup
3. Bandi Sravya - Server Setup, Backend Structure, Development, Configure MongoDB
4. Vijaya Raju Jakkamsetti - Database Connection, Schema & Models, Frontend Structure
5. Vankayala Jaya Chandra Srinivas - Steps for Execution, Demo Screenshots, Drive Link

2. Project Overview

Purpose & Goals

The BOOK STORE (BookVerse) application is a full-stack e-commerce web platform built utilizing the MERN stack (MongoDB, Express.js, React.js, and Node.js). It provides a responsive interface where users can search, browse, wishlist, and buy books. It handles role-based logins separating standard customers, sellers/vendors, and system administrators.

Key Product Features

- **Role-Based Registration & Login:** Tabbed credentials security separating customers, sellers, and system administrators using JWT.
- **Seller Catalog Controls:** Approved sellers can list new books, upload cover art files via Multer, modify details, and check order statuses.
- **Customer Shop View:** Customers browse book cards, search by text, filter by genre, add items to a local wishlist, and check out.
- **Fulfillment Tracker:** Sellers monitor orders containing their items and toggle shipping status (ontheway, delivered).
- **Admin Dashboard:** Administrators view system aggregates, inspect an SVG bar chart of database counts, approve sellers, edit user details, and cancel orders.

3. Technical Architecture

Frontend (React Client)

The frontend UI is developed with React.js using Vite. Global authentication contexts manage user sessions and tokens, and Bootstrap handles responsive viewport scaling.

Backend (Node.js & Express.js)

The backend API uses Node.js and Express.js styled under the Model-View-Controller (MVC) design pattern. Mongoose manages DB transactions, Multer handles image uploads, and Bcryptjs handles password encryption.

Database Schema (MongoDB)

- **Admins:** Stores administrator name, credentials, and self ID.
 - **Sellers:** Stores vendor name, credentials, and isApproved boolean state.
 - **Users:** Stores customer name, credentials, and timestamps.
 - **Books:** Stores title, author, genre, price, stock quantity, cover image path, and seller references.
 - **Orders:** Stores customer name, delivery address, booking date, delivery target, price, and status (ontheway, delivered).
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4. Setup & Setup Instructions

Prerequisites

- Node.js (v16+) and npm installer installed.
- A MongoDB Atlas cluster configured with network whitelisting enabled.

Installation

- 1. Extract/clone the repository folder structure.
- 2. In backend/ folder, create a .env file containing:

```
PORT=8000
MONGO_URI=mongodb+srv://23pa1a5761_db_user:AR15l56xvSCDDLmM@book-store.ouo6cb6.mongodb.net/bookverse?appName=Book-Store
JWT_SECRET=bookversesecrettokenkey12345
```

- 3. Run "npm install" inside both backend/ and frontend/ folders to download packages.

5. Folder Structure

- **frontend/src/App.jsx**: Coordinates browser routes.
 - **frontend/src/context/AuthContext.jsx**: Manages active roles and API connections.
 - **frontend/src/components/**: Layout components (Navbar.jsx, OrderModal.jsx).
 - **frontend/src/pages/**: Dashboard and catalogue pages.
 - **backend/server.js**: Core Node Express entry point.
 - **backend/seed.js**: Clear and populate MongoDB collections.
 - **backend/controllers/**: Express business logic controllers.
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6. Running the Application

Start the local development servers using two separate terminals:

Backend Service (Port 8000)

```
cd backend  
npm start
```

Frontend React App (Port 5173)

```
cd frontend  
npm run dev
```

7. API Documentation

- **POST /api/users/register**: Create customer account.
- **POST /api/users/login**: Verify customer credentials.
- **GET /api/users/books**: List books. Supports query filters (?search=text&genre=fiction).
- **POST /api/users/orders**: Checkout. Body: { bookId, customerName, address, price }.
- **POST /api/seller/books**: Add new product listing (Multipart file attachment).
- **PUT /api/admin/sellers/approve/:id**: Toggle vendor isApproved to true.

8. Authentication & Security

- **Hashing**: Passwords are encrypted with Bcryptjs using 10 salt rounds.
 - **JWT Bearer**: Authorizations are checked by parsing the Authorization Bearer header.
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10. Testing Strategy

- **Database Seed Check:** Seeder validates direct database clear and population.
- **Model Query Integrity:** `verify_backend.js` script queries Mongoose schema directly to verify record counts on Atlas.
- **Production Bundler Compile:** "npm run build" compiled with zero warnings.

11. Screenshots & Demo Drive Link

- **Full Demo Video Drive link:** <https://drive.google.com/drive/folders/1hORJ11561rBsZBxxFH1a3c4GGZ0JgF6y>

- **Seeded Atlas database verification output:**

```
MongoDB Connected: ac-6mn6yfbz-shard-00-00.ouo6cb6.mongodb.net
Admins found: 1 (Abhi)
Sellers found: 2 (Pravanshu - Approved, John Doe - Pending)
Users found: 3 (Ram, Akash, Siya)
Books found: 6 (The Alchemist, Atomic Habits, Rich Dad Poor Dad, etc.)
Orders found: 2 (Ram order - ontheway, Akash order - ontheway)
```

12. Known Issues

- Atlas Network whitelist requires 0.0.0.0/0 to allow global connections.
- Multer image uploads are limited to 5MB.

13. Future Enhancements

- Integrate Stripe checkout API for credit card transaction processing.
 - Incorporate email notifications via Nodemailer for low stock alerts.
-

UI Showcase: BookVerse Guest Home Page

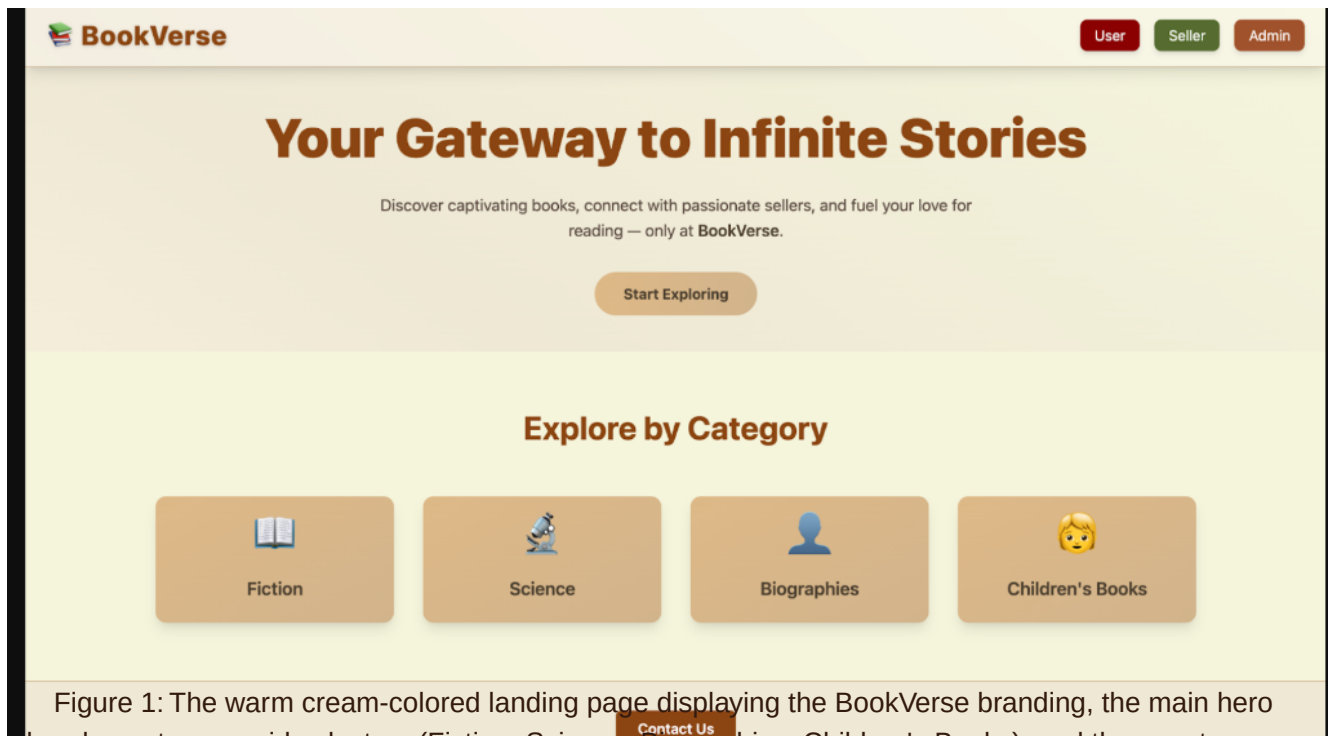


Figure 1: The warm cream-colored landing page displaying the BookVerse branding, the main hero header, category grid selectors (Fiction, Science, Biographies, Children's Books), and the guest access portals (User, Seller, Admin) in the top-right navbar.

UI Showcase: Administrator Dashboard



UI Showcase: Administrator Users List

BookStore (Admin)

HomeUsersSellersLogout(Abhi)

Users

Sl/No	User ID	Name	Email	Operation
1	68959952d9691a6325af12a3	Ram	ram@gmail.com	View Orders
2	68959db78e1cfa3b586c7437	Akash	Akash@gmail.com	View Orders
3	6895c91aa3d7833b2a94a410	Siya	Siya@gmail.com	View Orders

Figure 3: Detailed users management view within the admin dashboard listing Sl/No, User ID, Name, Email, and operations (Edit, Delete, and View Orders modal buttons).

UI Showcase: User Orders Modal Panel

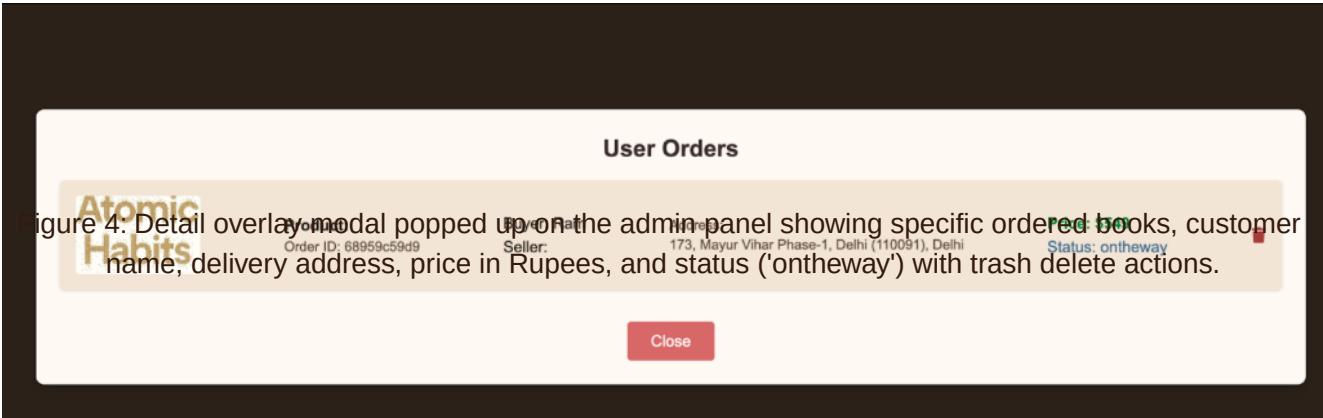


Figure 4. Detail overlay modal popped up on the admin panel showing specific ordered books, customer name, delivery address, price in Rupees, and status ('ontheway') with trash delete actions.

UI Showcase: Seller Dashboard

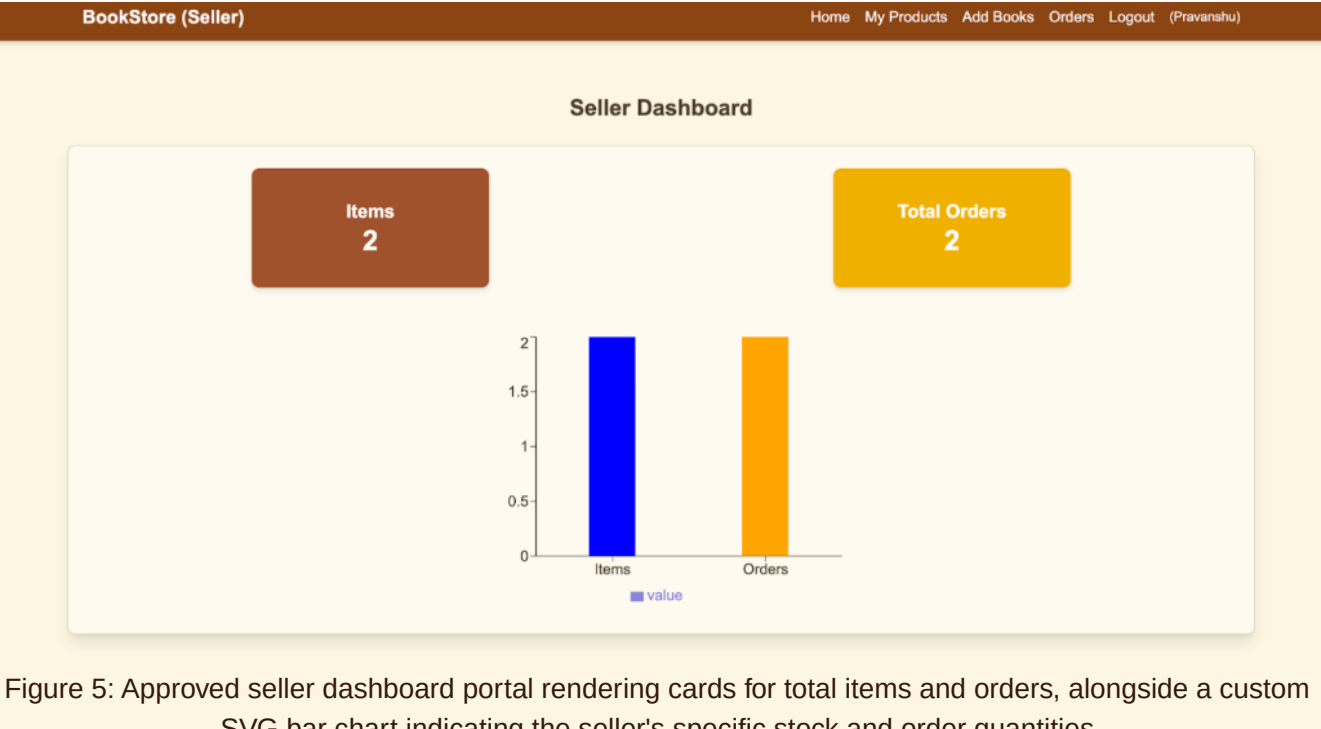


Figure 5: Approved seller dashboard portal rendering cards for total items and orders, alongside a custom SVG bar chart indicating the seller's specific stock and order quantities.

UI Showcase: Seller Book Catalog List

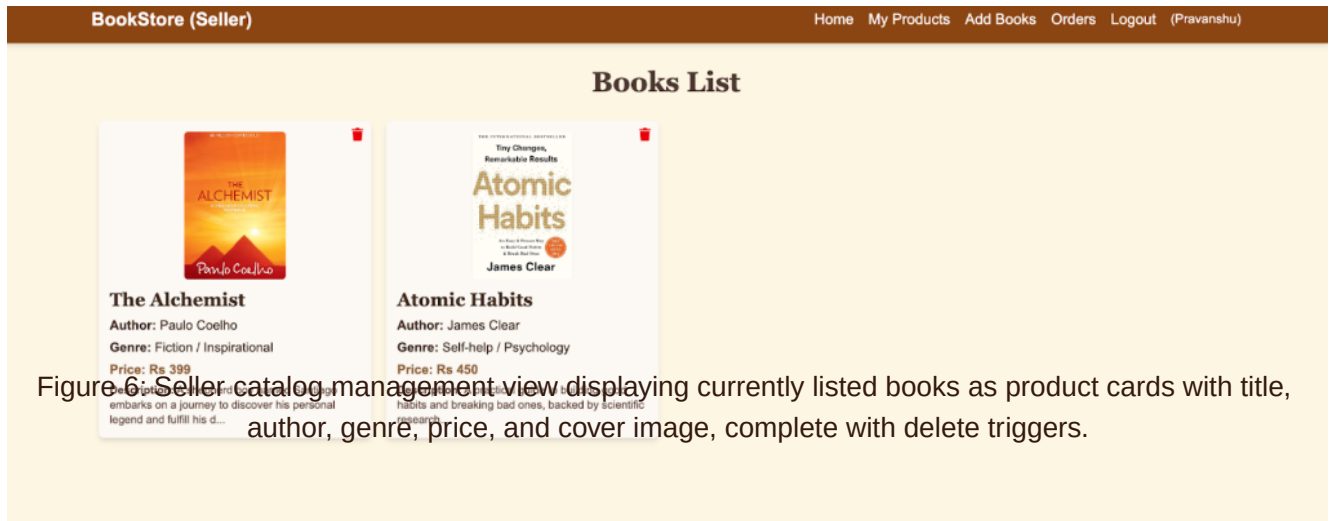
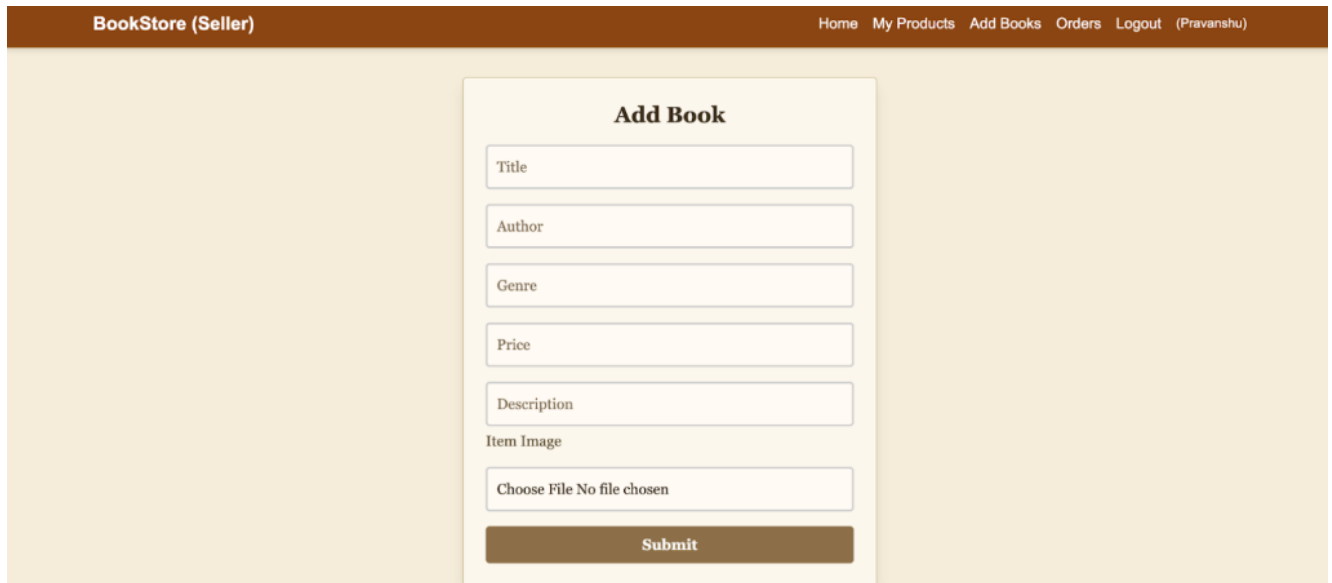


Figure 6: Seller catalog management view displaying currently listed books as product cards with title, author, genre, price, and cover image, complete with delete triggers.

UI Showcase: Seller Add Book Form



The image shows a web application interface for a seller to add a new book. The header is a dark brown bar with the text 'BookStore (Seller)' on the left and navigation links 'Home', 'My Products', 'Add Books', 'Orders', 'Logout', and '(Pravanshu)' on the right. The main content area has a light beige background. In the center is a white form titled 'Add Book'. The form contains the following elements: five text input fields labeled 'Title', 'Author', 'Genre', 'Price', and 'Description'; an 'Item Image' section with a file upload button labeled 'Choose File No file chosen'; and a dark brown 'Submit' button at the bottom.

BookStore (Seller) Home My Products Add Books Orders Logout (Pravanshu)

Add Book

Title

Author

Genre

Price

Description

Item Image

Choose File No file chosen

Submit

Figure 7: Product creation interface for sellers allowing inputs for Title, Author, Genre, Price, Description, and a cover image file upload.

UI Showcase: Seller Orders Log Sheet

BookStore (Seller)

HomeMy ProductsAdd BooksOrdersLogout (Pravanshu)

Orders

Atomic Habits

James Clear

Product Name

Order ID

Customer Name

Address

Booking Date

Delivery By

Warranty

Price

Status

68959c59d9

Ram

173, Mayur Vihar Phase-1, Delhi (110091), Delhi

2025-08-08

2025-08-13

1 year

₹549

ontheway

Book Cover

Product Name

Order ID

Customer Name

Address

Booking Date

Delivery By

Warranty

Price

Status

68959deb8e

Akash

Balaji Medical, Datia (475661), Madhya Pradesh

2025-08-08

2025-08-13

1 year

₹498

ontheway

Figure 8: Seller orders log sheet showing details of orders placed for their books, booking dates, delivery estimations, price, and status controllers.

Figure 8: Seller orders log sheet showing details of orders placed for their books, booking dates, delivery estimations, price, and status controllers.

UI Showcase: Customer Home Catalog Page

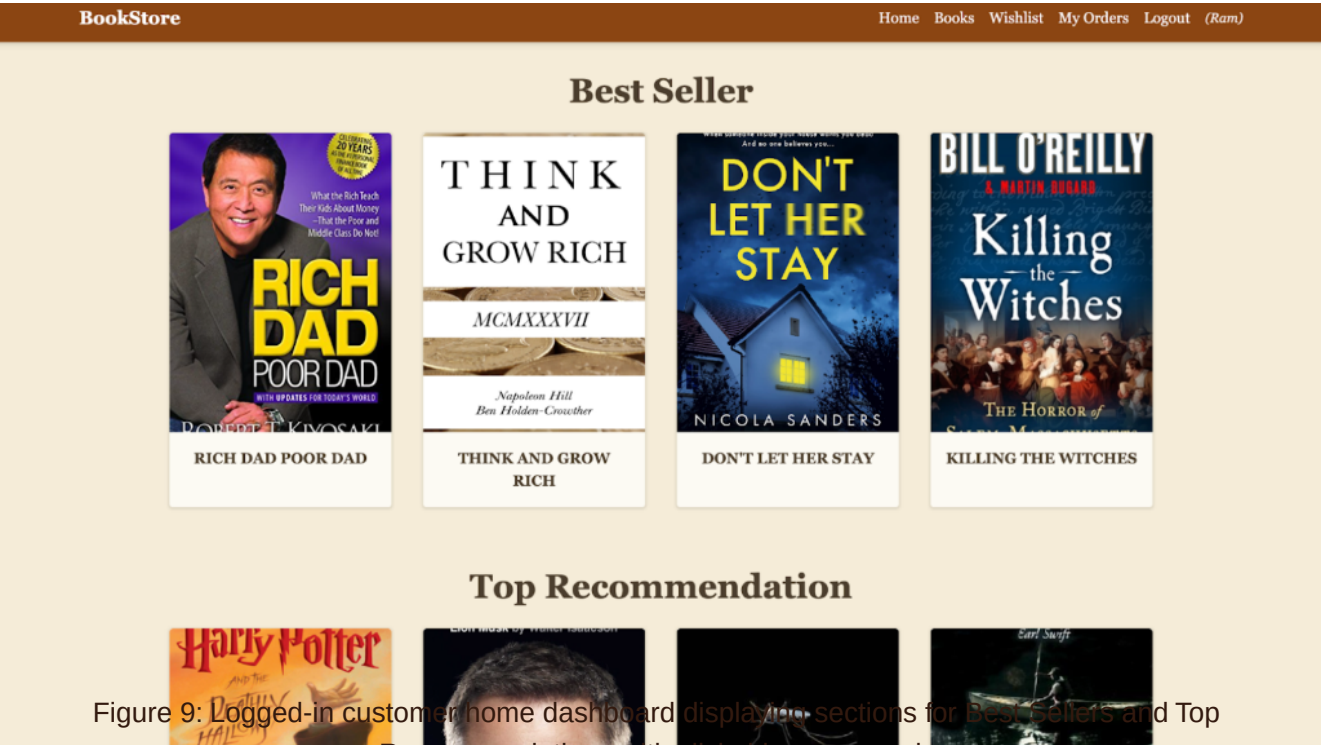


Figure 9: Logged-in customer home dashboard displaying sections for Best Sellers and Top Recommendations with clickable cover cards.

UI Showcase: Customer Full Books List Grid

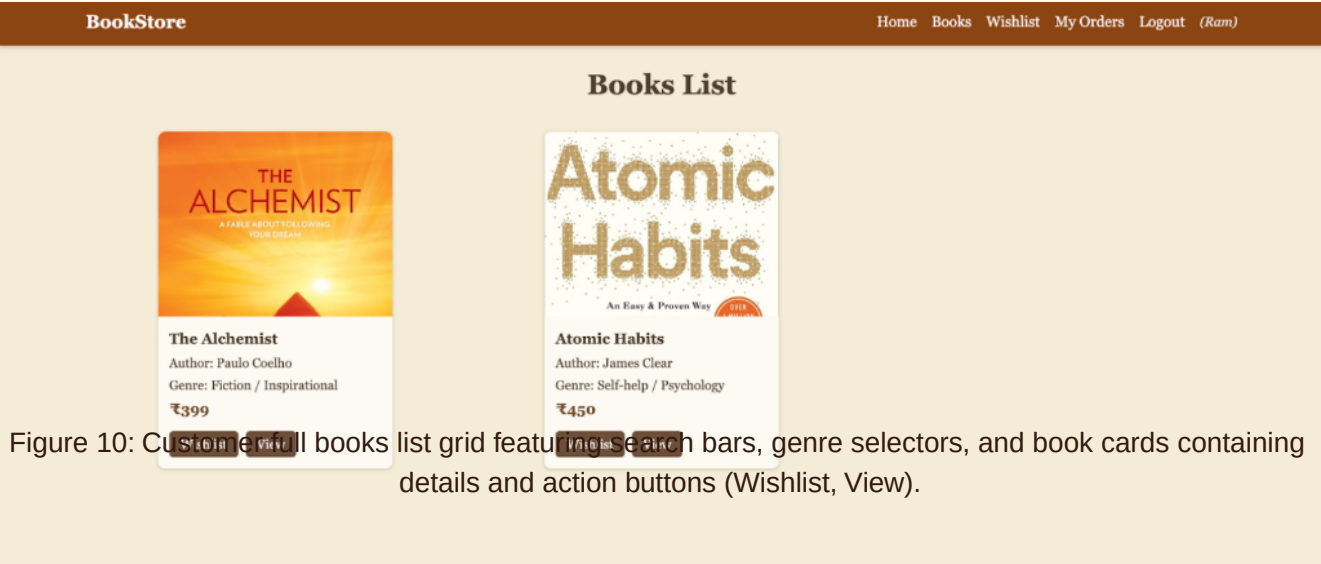


Figure 10: Customer Full books list grid featuring search bars, genre selectors, and book cards containing details and action buttons (Wishlist, View).

UI Showcase: Customer Book Details View

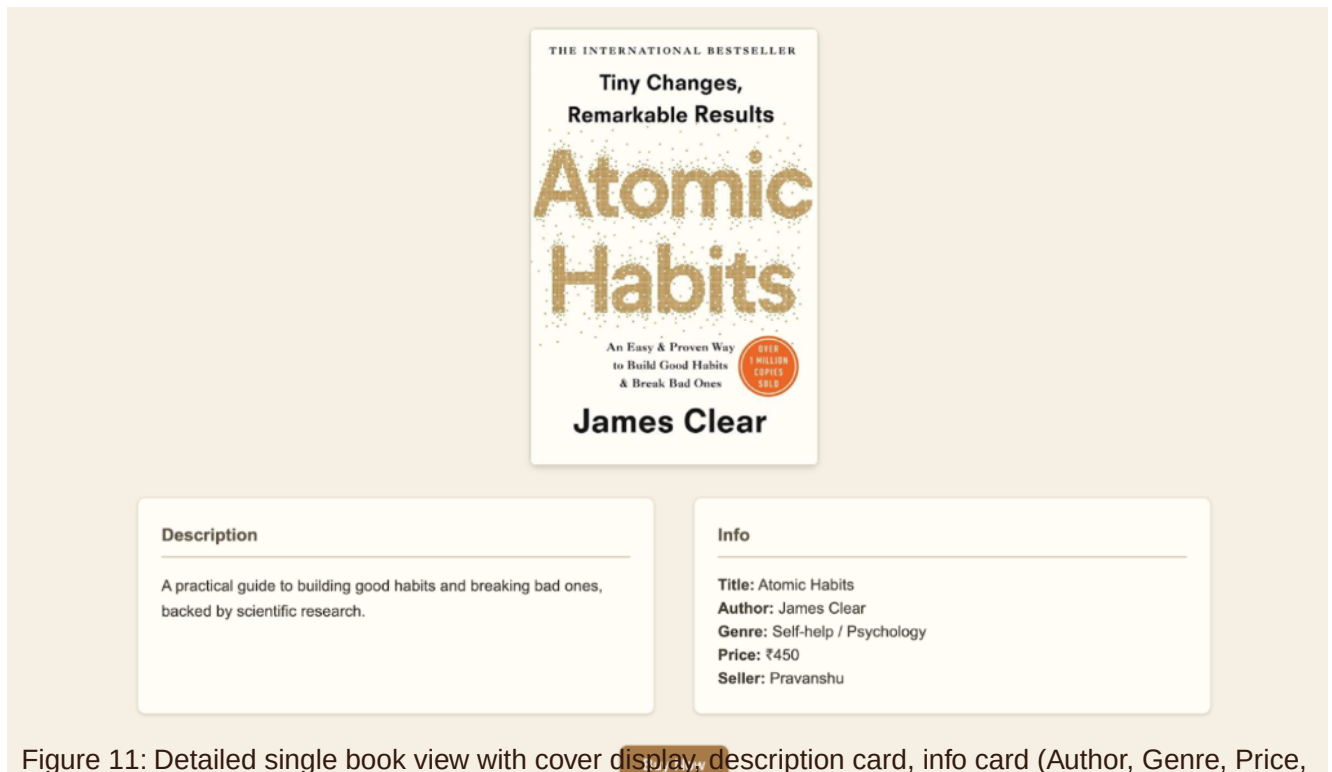


Figure 11: Detailed single book view with cover display, description card, info card (Author, Genre, Price, Seller), and the Buy Now checkout button.

UI Showcase: Customer My Orders Tracker

BookStore

[Home](#) [Books](#) [Wishlist](#) [My Orders](#) [Logout](#) (Ram)

My Orders



Product Name

Atomic Habits - 59c5

Order ID

68959c59d9

Address

173, Mayur Vihar Phase-1, Delhi (110091), Delhi

Seller

Booking Date

2025-08-08

Delivery By

2025-08-13

Price

₹549

Status

ontheway

Contact Us

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Figure 12: Customer order tracker list page displaying historical order cards, tracking statuses, and the BookVerse contact footer.

